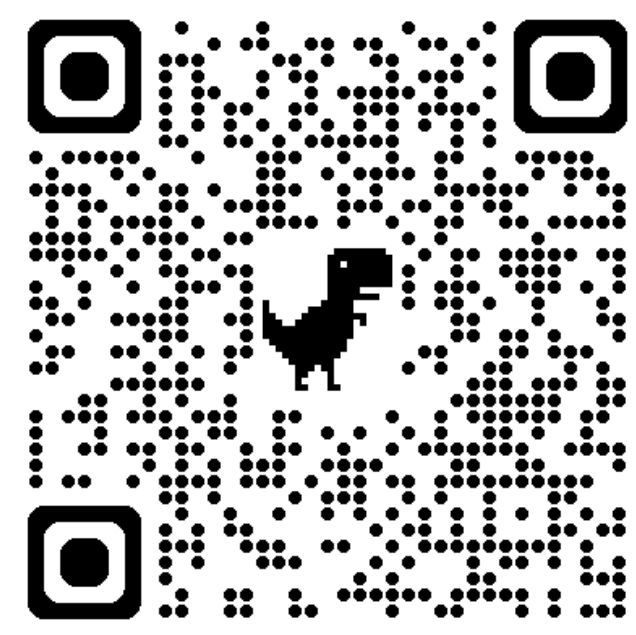
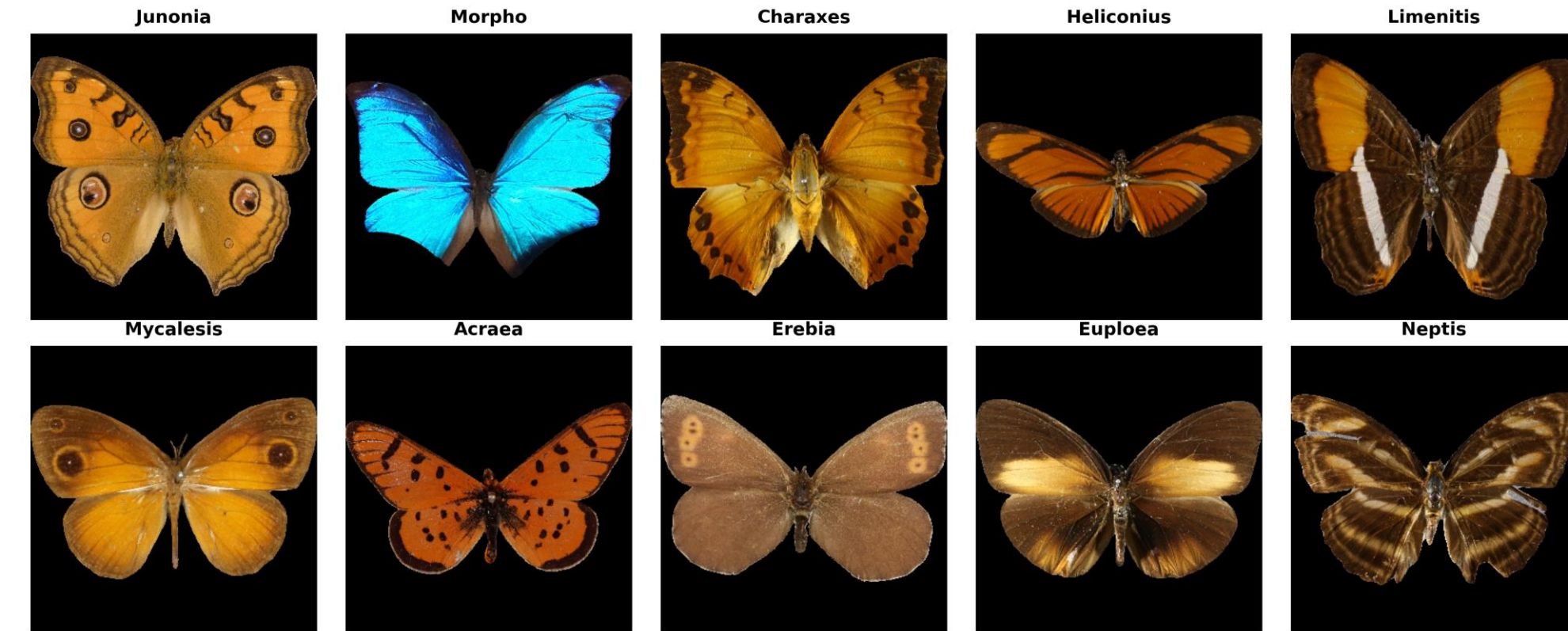




1. Motivation : Collection-Scale Bottleneck?



Digitization Acceleration : Biological image archives are expanding far faster than manual expert curation can scale.



Repository Noise : Historical collections contain unresolved boundaries, labeling errors, and subtle population structures.



Supervised Trap: Supervised classifiers force images into predefined species labels, inheriting training label noise and failing on novel or variable morphs.

Proposed Approach: Visual clustering offers a practical triage tool for large digitized natural history collections

2. Dataset: Why Butterflies? A Rich Vision Benchmark

Biological significance

- Wing patterns encode mimicry, polymorphism, geography, seasonality, and divergence.
- Species labels and visual phenotypes are related but non-equivalent.
- Mismatches can guide expert review and collection triage.

Computer vision significance

- Fine-grained differences and within-species variation make recognition hard.
- Open-set structure exposes limits of fixed-label classification.
- Long-tailed sampling reflects real biodiversity archives.
- Supports clustering, retrieval, anomaly detection, and



Butterflies provide a rare testbed where visual similarity, taxonomy, and biological process interact—but do not always agree.

3. Benchmark: Species-Capped 1-50 Sampling

230,000+

SOURCE IMAGES

189,541

DORSAL ARCHIVE

11,821

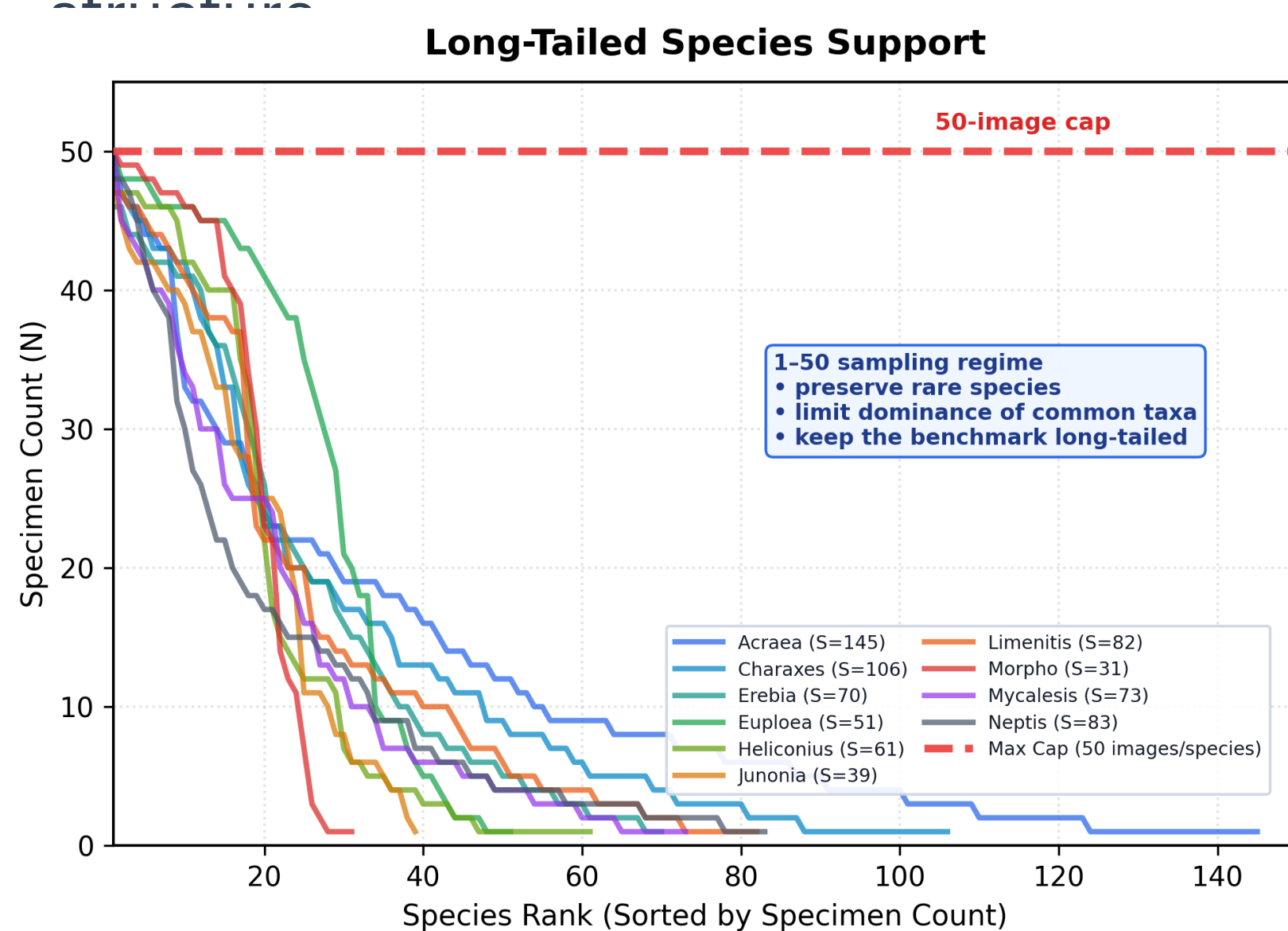
BENCHMARK IMAGES

741

RECOGNIZED SPECIES

Key Design : 1-50 images/species

- Retain rare taxa • Cap dominant species • Preserve long-tailed



Genus	S	N	N/S
Junonia	39	919	23.56
Morpho	31	943	30.42
Charaxes	106	1,403	13.24
Heliconius	61	1,020	16.72
Limenitis	82	1,246	15.20
Mycalesis	73	1,030	14.11
Acraea	145	1,675	11.55
Erebia	70	1,206	17.23
Euploea	51	1,378	27.02
Neptis	83	1,001	12.06

Species-capped sampling retains rare species in the benchmark while preventing abundant taxa from dominating evaluation.

7. Conclusion: What this work enables

Taxonomy-free organization of large specimen archives by visual phenotype.

Automated triage of fragmented taxa, mixed clusters, atypical specimens, and label inconsistencies.

A reproducible benchmark for evaluating foundation-model representations in biological collections.

Self-supervised vision can recover meaningful phenotype structure—but phenotype and taxonomy are not equivalent.

4. Methodology: Taxonomy-Free Benchmarking Pipeline

1 Images

11,821 Dorsal specimens



2 DINOv3

zero-shot embeddings



3 Representation

PCA · t-SNE · UMAP



4 Clustering

Ward · HDBSCAN



5 Evaluation

ARI · Silhouette



Zero Shot Visual Structuring

No taxonomic labels are used during feature extraction, representation reduction, or clustering.

Post-hoc taxonomic audit

Species names are introduced only afterward to measure whether visual structure aligns with taxonomy.

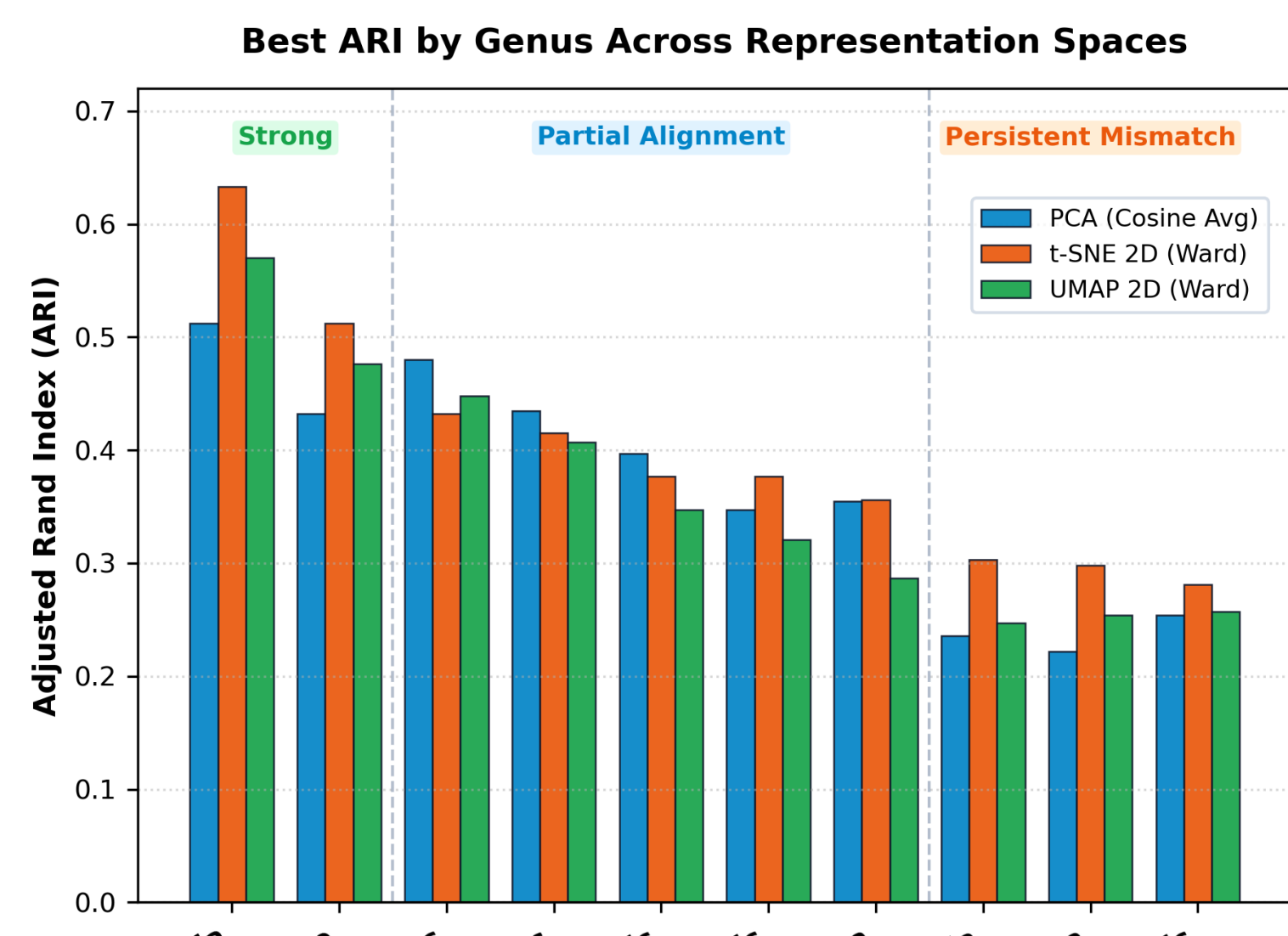
Extrinsic vs. intrinsic metrics

ARI quantify taxonomic agreement; Silhouette measures visual coherence.

The model is not trained to predict taxonomy; taxonomy is used only afterward as an audit signal.

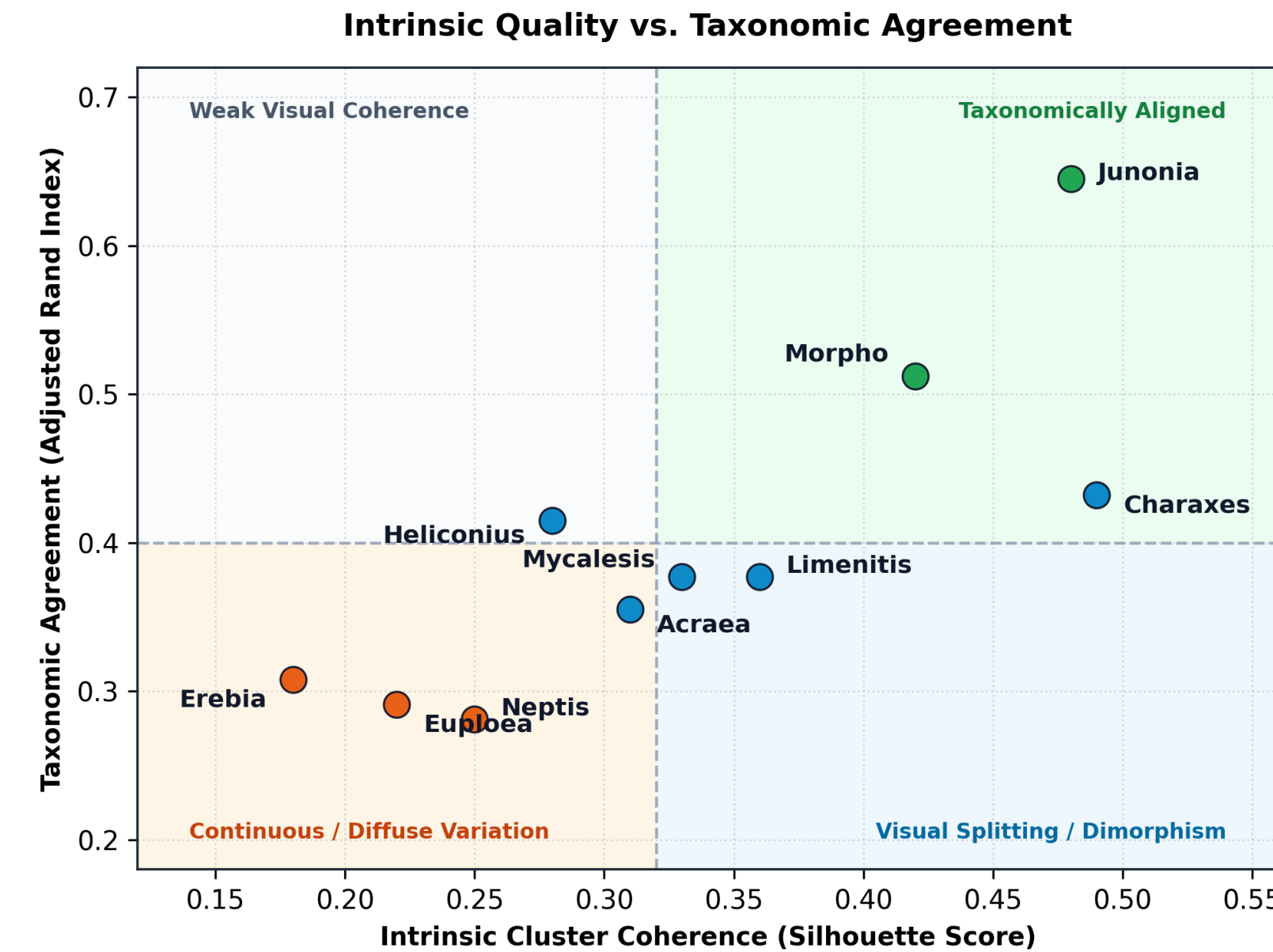
5. Quantitative Results

5.1 Phenotype–Taxonomy Alignment



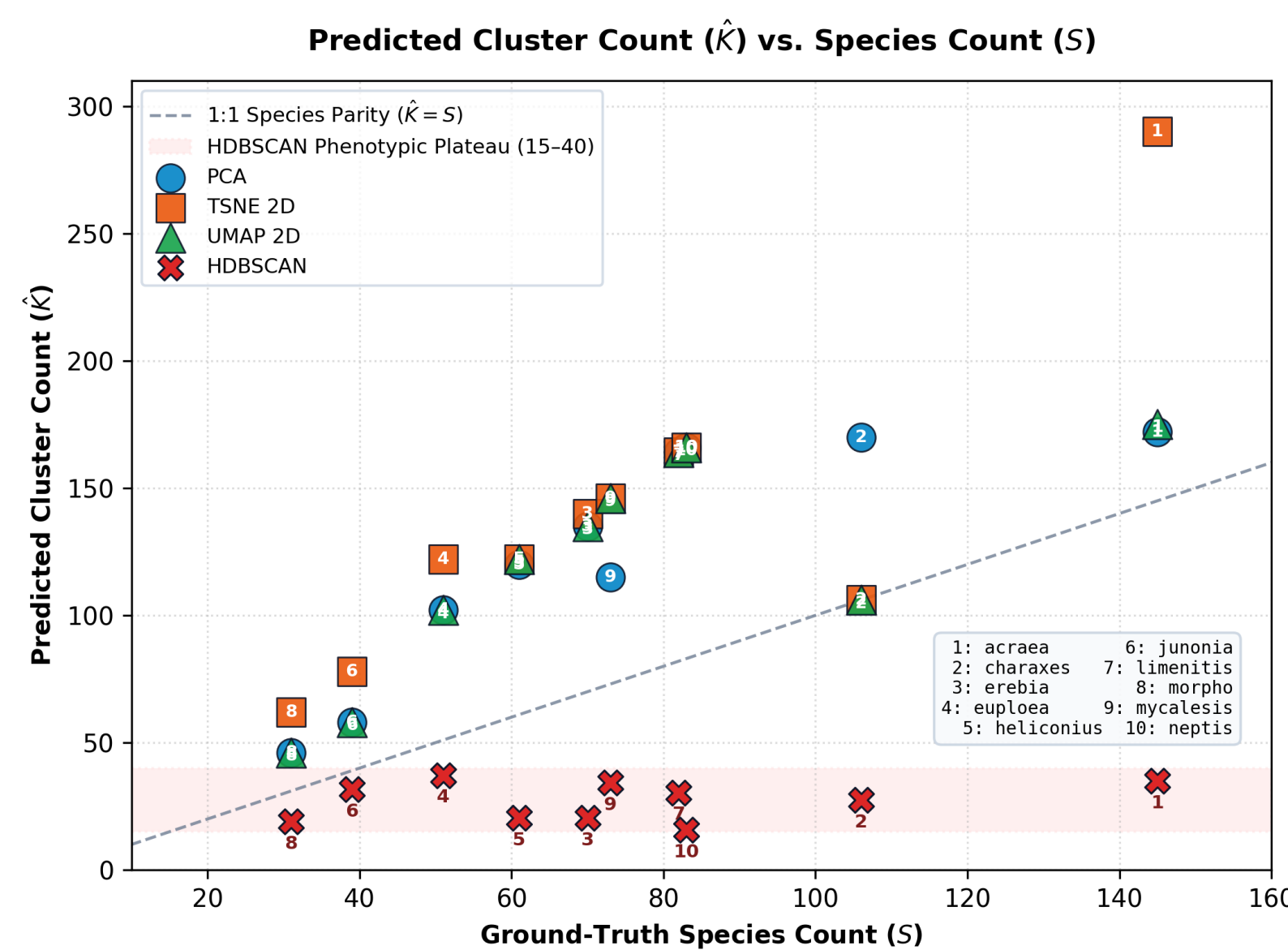
Phenotype–taxonomy alignment is driven by genus-specific biology.

5.2 Visual Coherence Does Not Equal Taxonomic Agreement



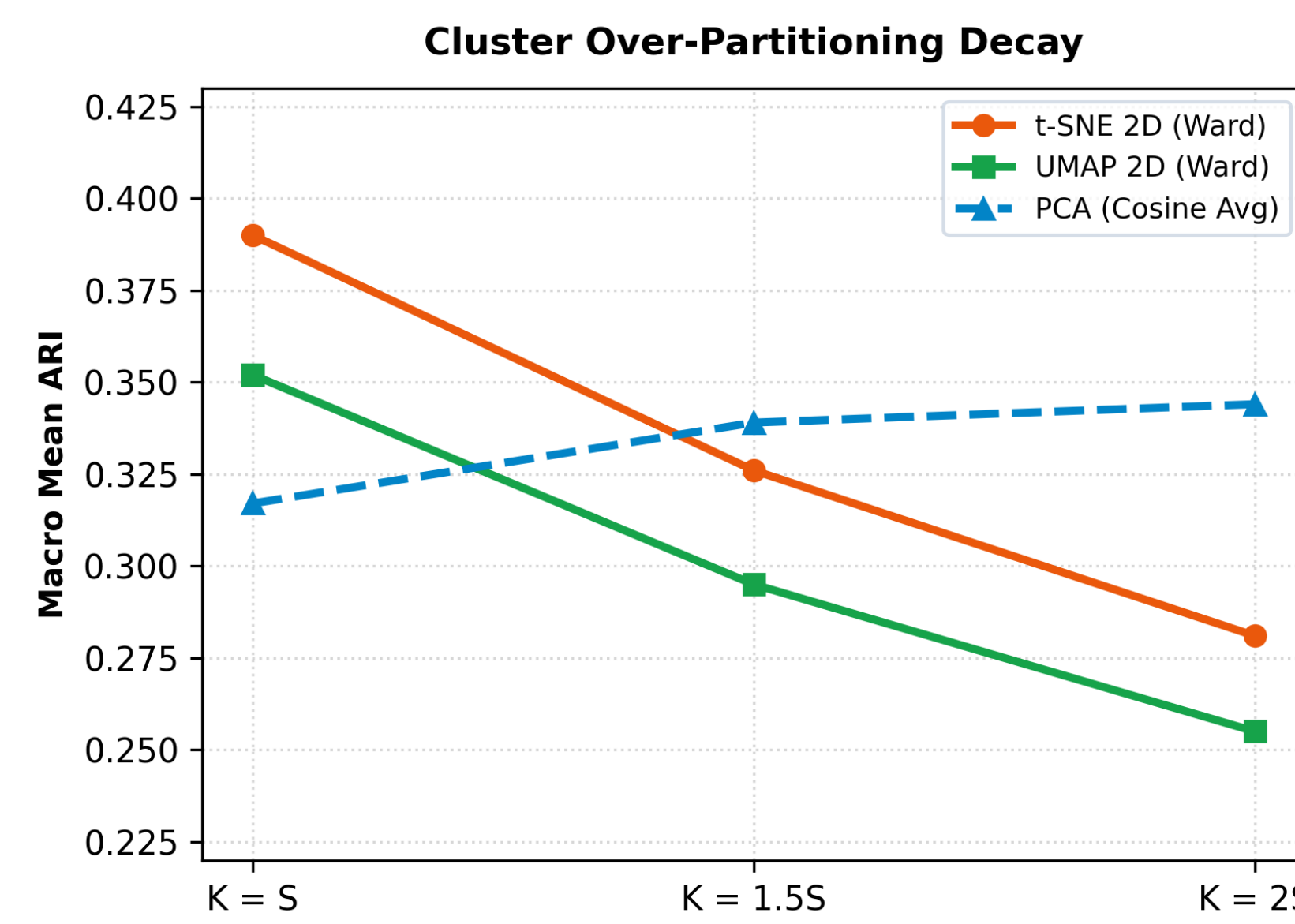
Phenotypic organization and taxonomy are related—but non-equivalent.

5.3 Dense Phenotypic Modes Are Coarser Than Species Richness



Dominant phenotypic structure is often coarser than species-level taxonomy.

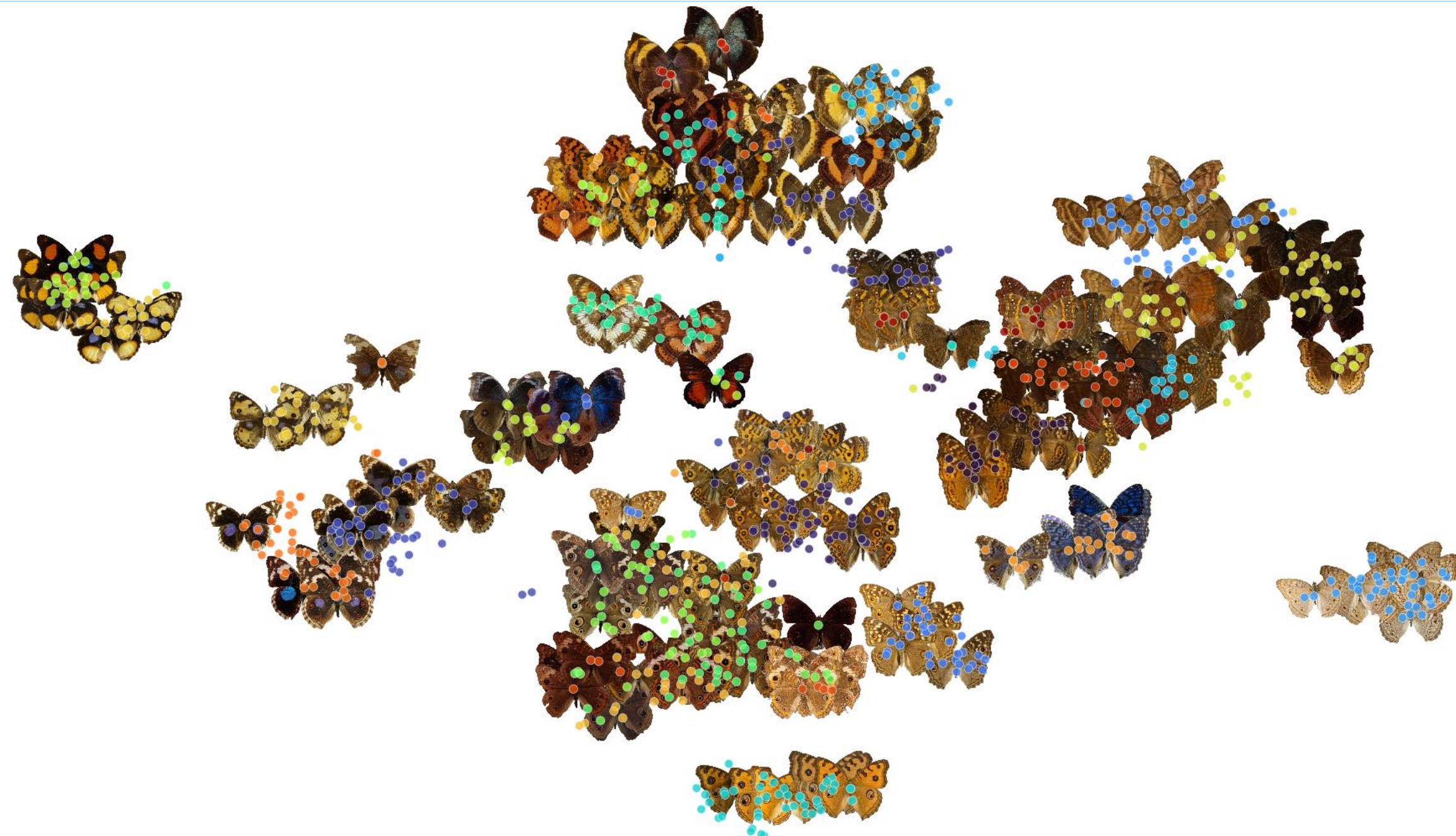
5.4 Nonlinear Manifolds Reveal Fine Scale Phenotypic Structure



Finer partitions expose within-species phenotypic substructure.

6. Qualitative Results: Alignment Regimes

6.1 Strong Alignment : Junonia



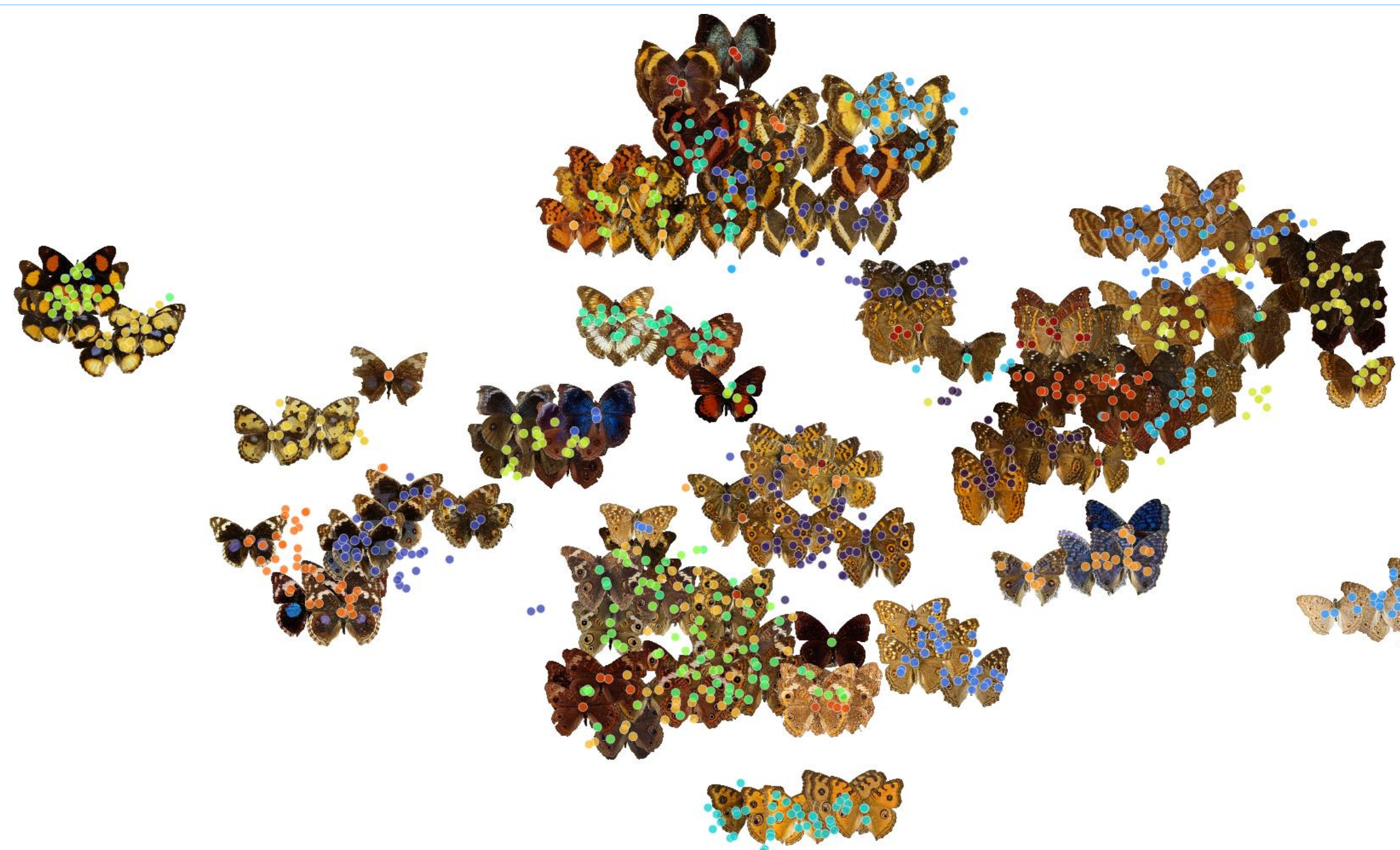
Junonia: Discrete phenotypic neighborhoods broadly track species.

6.2 Partial Alignment : Charaxes



Charaxes: Coherent phenotypic modes fragment nominal species.

6.3 Persistent Mismatch : Erebia



Erebia: Continuous phenotypic variation weakens species separation.

9. Collaboration Opportunities

- Geometric ML & graph representation learning
- Open-set retrieval & anomaly detection
- Vision foundation models for biodiversity
- Cross-collection / multimodal biological AI

Seeking collaborators across computer vision, geometric ML, biodiversity informatics, and natural-history collections.

8. Future Direction: From Visual Phenotype to Knowledge-Grounded Vision

Phylogeny

Evolutionary structure

Geography & Time

Spatial / Seasonal

Ecology

Host and environment

Provenance

Digitization Effects

DINOv3 / BioCLIP
Embeddings

Knowledge-Grounded Vision
Foundation-model alignment

Open-Set Retrieval
Collection Triage

Align visual embeddings with relational biological structure for Open-Set Retrieval and Collection Triage.